AISE 4010 — Deep Learning for Time Series

COURSE PROJECT – Fall 2026

• one evolving Jupyter notebook • three in-lab demonstrations

Project weight	45% of final course grade	**Group size**	4 students (may change based on enrolment number)
In-lab marking	Three stages: Labs 6, 9 and 11	**Presentation**	Live notebook and outputs demo

1. Project objective

Build and evaluate a complete deep-learning pipeline on one approved public time-series dataset. The project develops alongside the labs. Each group applies the week's method to its own data, records the work in one evolving Jupyter notebook, and demonstrates evidence at three marked checkpoints.

The grade rewards a correct implementation, appropriateness of methodology, correct forecasting protocol, fair model comparison, reproducibility, clear analysis, interpretation of results, and demonstrated understanding of the work—not the smallest raw error or the most complicated architecture. Live questions or small code modifications may be used to assess individual understanding.

Submission contract

All executable and analytical project work must be in AISE4010_Project_GroupNN.ipynb. The group also submits a GitHub repository and a four-page full report. The final Lab 11 demonstration is the complete project presentation through the executed notebook and its outputs.

1.1 Required core task

- Preprocess and split the dataset appropriately, define the target and lookback horizon before modelling.
- Compare traditional baselines with MLP, CNN, recurrent, TCN and Transformer-family models under the same split and metrics.
- Watch carefully for data leakage.
- Select hyperparameters using validation data only; the test set remains untouched until the final stage.
- Explain when the final model succeeds, when it fails, and what evidence supports the conclusions.

2. How the project follows the labs

Reserve time after each tutorial/lab assignment to transfer that method to the project dataset. You have the chance to conveniently work with your group members during the lab hours and get help from the TAs for any issues/questions. The notebook should grow in the same order as the course.

Lab	Activity	Required progress
Lab 1 (week of Sep 14)	Tutorial	Claim a dataset; exploratory analysis, define target, frequency and horizon.
Lab 2 (week of Sep 21)	Tutorial	Last chance to claim a dataset (if you haven't done in Lab 1). Clean/resample; chronological split; traditional baselines.
Lab 3 (week of Sep 28)	Tutorial/in-lab Assignment	Create windows; train MLP; inspect learning curves and check overfitting.
Lab 4 (week of Oct 5)	No Lab	-
Lab 5 (week of Oct 19)	Tutorial/in-lab Assignment	Add CNN and tune using grid search. Work on project to be prepared for next week's demo
Lab 6 (week of Oct 26)	**STAGE 1 DEMO**	Implement traditional baselines, MLP, CNN; tune the best model with grid search; compare test results and computational cost.
Lab 7 (week of Nov 2)	Tutorial	Add RNN/LSTM/GRU.
Lab 8 (week of Nov 9)	Tutorial/in-lab Assignment	Add TCN model.
Lab 9 (week of Nov 16)	**STAGE 2 DEMO**	Implement RNN family and TCN; tune the best model with grid search, compare test results and computational cost; optional Jetson deployment.
Lab 10 (week of Nov 23)	Tutorial/in-lab	Add Transformer model and HPO.
Lab 11 (week of Nov 30)	**STAGE 3 FINAL DEMO**	Implement Transformer and tune one promising model with a documented validation-only budget. Time/frequency (if applicable), fairness/uncertainty analysis (optional), failure analysis, and optional Jetson deployment. Present the complete project end-to-end through code and outputs.

After the last lab (Lab 11): Polish notebook; write four-page report; submit final GitHub and report.

Fall Reading week: Oct 10-18

2.1 Evidence discipline

- Before each demo, push the current notebook and record the GitHub commit hash. Marking is based on that checkpoint plus the incremental Git activities for individual group members, as well as live demo.
- Long training does not need to be repeated live. Keep outputs, checkpoints, logs and a lightweight smoke-test cell so the TA can verify the pipeline.
- Every group member must be able to explain the data, split, models, results and code decisions.

3. Dataset selection and common protocol

Choose one unclaimed dataset from the verified pool and register in the lab with TAs (Lab 1 or if you couldn't do it, in Lab 2, which is your **last chance**). The separate CSV catalogue is the authoritative source for links, target names, sizes, resampling and dataset-specific cautions.

Rule	Required implementation
Split	Chronological split (e.g., 60% train / 20% validation / 20% test). Never shuffle time globally.
Preprocessing	Fit scalers, imputers, encoders and feature-selection decisions on training data only.
Inputs	Historical target values and future-known calendar variables are allowed. Observed weather or other contemporaneous measurements may be used only as lagged inputs unless future availability is justified.
Metrics	Report multiple metrics (e.g., MAE, RMSE, Accuracy, F1-Score (if applicable), etc., and improvement over baseline (%)).
Fair comparison	Use identical settings (e.g., splits, windows, augmentation, metrics and test instances) across models for fair comparison and record parameters, training time and seed.
Hyperparameter Optimization (HPO)	Do the HPO for at least the best selected model/models and report the configurations and results. They should be optimized on the validation set only.

3.1 Verified data pool

#	Dataset	Access link
1	Household Electric Power	https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption
2	Appliances Energy Prediction	https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction
3	Tetouan City Power	https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city
4	Open Power System Data	https://data.open-power-system-data.org/time_series/
5	Australian Electricity Demand	https://zenodo.org/records/4659726
6	Metro Interstate Traffic	https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume
7	Seoul Bike Sharing Demand	https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand
8	Bike Sharing (Capital)	https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset
9	Melbourne Pedestrian Counts	https://zenodo.org/records/3898350
10	Beijing Multi-Site Air Quality	https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data
11	Air Quality (Italy)	https://archive.ics.uci.edu/dataset/360/air+quality
12	Jena Climate 2009–2016	https://storage.googleapis.com/tensorflow/tf-keras-datasets/jena_climate_2009_2016.csv.zip
13	Saugeen River Flow	https://zenodo.org/records/4656058
14	Human Activity Recognition	https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones
15	Room Occupancy Estimation	https://archive.ics.uci.edu/dataset/864/room+occupancy+estimation
16	NN5 — ATM cash withdrawals	https://zenodo.org/records/3889740
17	London Smart Meters	https://zenodo.org/records/4656091
18	PEMS08 — traffic flow	https://raw.githubusercontent.com/guoshnBJTU/ASTGNN/main/data/PEMS08/PEMS08.npz

#	Dataset	Access link
19	Traffic — SF Bay Area	https://raw.githubusercontent.com/laiguokun/multivariate-time-series-data/master/traffic/traffic.txt.gz
20	Electricity — 321 clients	https://raw.githubusercontent.com/laiguokun/multivariate-time-series-data/master/electricity/electricity.txt.gz
21	ETTh2 — transformer temp.	https://raw.githubusercontent.com/zhouhaoyi/ETDataset/main/ETT-small/ETTh2.csv
22	Solar — 137 plants, Alabama	https://raw.githubusercontent.com/laiguokun/multivariate-time-series-data/master/solar-energy/solar_AL.txt.gz
23	Beijing PM2.5	https://raw.githubusercontent.com/jbrownlee/Datasets/master/pollution.csv
24	FordA — engine fault	https://raw.githubusercontent.com/hfawaz/cd-diagram/master/FordA/FordA_TRAIN.tsv
25	MotionSense — phone IMU	https://raw.githubusercontent.com/mmalekzadeh/motion-sense/master/data/A_DeviceMotion_data.zip
26	M4 Hourly	https://raw.githubusercontent.com/Mcompetitions/M4-methods/master/Dataset/Train/Hourly-train.csv

*Source metadata and official landing pages were checked. Each group must successfully load its assigned target and show the timestamp audit by Lab 2. A broken download must be reported immediately, **not replaced silently**.*

4. Stage 1 — Data, baselines, MLP

Lab 6 • 5%

The notebook must be executed with outputs visible.

- Dataset source, exact target, unit, native frequency, resampling rule, coverage, gaps, duplicates, missingness and at least two informative exploratory figures.
- Chronological split diagram/table; scaler and imputer; explicit window shapes.
- At least one simple non-neural network baseline covered in the labs.
- MLP and CNN trained with the same inputs, horizon, loss, metrics and validation protocol as traditional baselines.
- Comparison table with metrics, parameters, training time and baseline improvement.
- Grid search configuration and results for best model.
- One concrete conclusion and one unresolved risk to investigate next.

Stage 1 marking (5 points)

Criterion	Pts
Data preprocessing and exploratory analysis	**0.5**
Data split pipeline	**0.5**
Baselines	**1.0**
MLP, CNN	**1.5**
Grid search	**0.5**
Live explanation	**1.0**

5. Stage 2 — Sequence models (Recurrent family), CNN, TCN, Grid-Search Hyperparameter Optimization (HPO)

Lab 9 • 10%

Expand the comparison, tune one model responsibly and select a finalist using validation evidence only.

- Add RNN, LSTM/GRU, and TCN models; keep baselines, CNN and MLP in the same comparison.
- Use the same split, windows and metrics for all models. Report parameter count and training and inference time.
- Define the HPO search space, budget, objective and stopping rule before showing results.
- Show an HPO trials table/plot and select one final model using evidence, not a preferred architecture name.
- Checkpoint the selected model, configuration and preprocessing state for the final stage.

Stage 2 marking (10 points)

Criterion	Pts
Models implementations	4.0
Controlled comparison	2.0
HPO evidence	1.0
Model selection	1.5
Live explanation	1.5

6. Stage 3 — Final full-project demonstration

Lab 11 • 15% • This is the full project presentation

Present the entire project from problem definition to Transformer modelling, proper HPO, any complementary analysis (such as time-frequency, fairness/uncertainty analysis (optional), deployment (optional), etc., if applicable), and final conclusions through the executed Jupyter notebook and code repository.

6.1 Required presentation sequence

1. Problem and data: source, target, horizon, operational interpretation, audit and representative plots.
2. Evaluation contract: chronological split, window alignment, leakage controls, baselines and metrics.
3. Model development: MLP, CNN, recurrent model, TCN and Transformer-family model under one controlled protocol.
4. HPO and selection: validation evidence, trade-offs, selected configuration and why it won.
5. Final test: load the locked finalist, evaluate the untouched test set once, and compare every model and baseline in one final table.
6. Additional analysis: time/frequency insight, fairness analysis (optional), and at least one failure regime.
7. Reliability and limits: uncertainty or empirical error bands, robustness/sensitivity evidence, deployment limitations and next steps.
8. Hardware deployment (optional)
9. Reproducibility: GitHub commit, environment, run order, saved model and a working smoke test.

6.2 Figure and table traceability

Every figure, table and quantitative claim in the four-page report must:

- be generated by a clearly labelled notebook cell and remain visible in the submitted notebook output;
- use the same caption/identifier in the report and notebook (for example, Fig. 2 or Table 1); and
- be shown and explained during the final demo. A report-only figure or externally assembled result is not acceptable, except for clarification purposes, such as pipeline/framework figures.

The TA may ask the group to rerun any lightweight cell, change a displayed example, inspect an intermediate object, or explain any result. Training a full model live is not required.

Stage 3 marking (15 points)

Criterion	Pts	Evidence
All model implementations (MLP, CNN, recurrent model, TCN and Transformer), and end-to-end narrative	**6.0**	Complete modelling, coherent end-to-end project in the notebook.
Final test comparisons	**3.0**	Test evidence, correct metrics and interpretation.
HPO evidence	**2.0**	Validation evidence, selected configuration and why it won.
Advanced analysis	**1.0**	Time/frequency insight, fairness/uncertainty analysis (optional), and failure-regime insight.
Reliability + limitations	**1.0**	Robustness evidence and limitations.
Live Explanation & Reproducibility	**2.0**	All members defend code/results; repository can be checked.

6.3 Demo rules

- Use the final GitHub commit and one executed notebook. No slide deck, prerecorded narration or separate figure file is graded as the presentation.
- All members must attend and speak. The instructor may direct questions to any member.
- Cache data, model weights and essential outputs before the lab. Internet or training-time problems do not remove the evidence requirement.

7. Final GitHub repository - Due date: Dec 9, 2026 at 11:59 PM.

Repository Submission 1% • Due with the report

Submit the repository URL and final commit hash. The instructor/TA must have access before the deadline.

7.1 Required repository contents

Path	Requirement
AISE4010_Project_GroupNN.ipynb	The single authoritative executable artifact; run top-to-bottom with final outputs visible.
README.md	Problem, dataset/source, setup, data placement/download, exact run order, expected runtime and final commit.
requirements.txt or environment.yml	Pinned or bounded dependencies sufficient to reproduce the notebook.
data/README.md	Only instructions or small permitted data. Do not commit large, restricted or licensed raw files.
models/ or artifacts/	Optional small checkpoints/results if GitHub limits and dataset/model licences allow.

- All executable modelling, analysis and figure-generation code must be in the Jupyter notebook. Do not move core work into .py scripts.

- Use relative paths; never commit credentials, access tokens, private data, absolute local paths or generated caches.
- Use meaningful commits across the term. **A last-minute code dump is not evidence of the staged process,** and you will lose Stage's marks.
- Group members' commits will be tracked for individual marking.
- The final notebook must be restarted and run top-to-bottom before submission. Keep outputs; remove dead experiments and secret values.

8. Four-page full report (4 pages) - Due date: Dec 9, 2026 at 11:59 PM.

Report Submission 10% • Due with the repository

Groups should submit a final project report of a **maximum of four pages** total in IEEE two-column format, including an abstract, introduction and related work, dataset description, methodology details, experimental results, conclusion and limitation discussion, figures, tables, captions, references, and a description of group members' contributions (what part of the project each group member worked on). The templates for MS Word/LaTeX markups can be found at this link: Report Template.

The notebook may contain additional diagnostic evidence, but the report must stand on its own within four pages and every included exhibit must be demonstrated in Lab 11.

9. Project grading summary

Component	Weight
Stage 1 in-lab demo	**5%**
Stage 2 in-lab demo	**10%**
Stage 3 final full-project demo	**15%**
Final GitHub repository	**1%**
Four-page full report	**14%**
TOTAL	**45%**

9.1 Individual accountability

The project is completed in groups of four, but equal individual marks are not automatic. The instructor may adjust an individual's project mark using:

- answers during any of the three demos;
- the ability to explain any major preprocessing, model, metric, figure or result;
- GitHub history and a concise contribution log in the notebook; and
- documented peer/instructor evidence of substantially unequal participation.

A member who cannot explain the group's work may receive a lower individual mark even when the code runs.

9.2 Academic integrity and tool use

- Cite datasets, libraries, papers, tutorials and reused code. Follow the course policy for generative-AI tools and disclose substantial assistance.
- Published solutions may be read for context, but copying a solution notebook, result table or analysis without attribution and independent understanding is unacceptable.

- Do not fabricate runs, cherry-pick test windows, alter the test set after viewing results, or present externally created figures as notebook output.

10. Optional Jetson deployment bonus

Subject to device availability and instructor approval, a group may earn up to +2 project percentage points for a reproducible on-device deployment of the locked final model or a justified lightweight alternative. Show latency/resource measurements, numerical agreement or explained differences, and an edge-suitability discussion. Bonus work cannot replace any required evidence and remains subject to the course-grade cap.